

Simulating MITRE ATT&CK with Invoke-Atomic Red Team in Wazuh

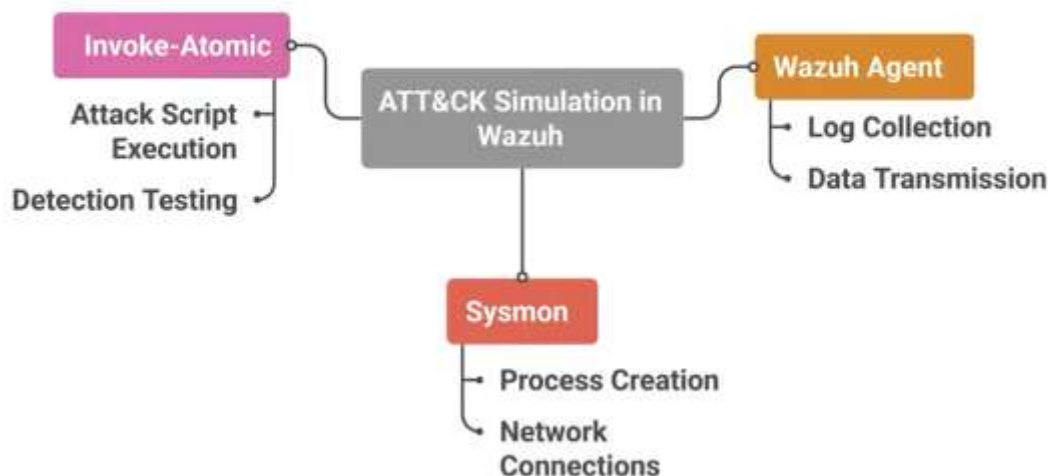
Simulating an attack with Invoke Atomic in Wazuh is a powerful method to test detection rules.

We will use **Invoke Atomic Red Team**, which simulates MITRE ATT&CK framework-based attacks.

1- The three main components in this process to simulate MITRE ATT&CK in wazuh:

- **Wazuh agent:** The Wazuh agent has a messenger role because it **collects logs** from the agent machine and **sends them** (Data transmission) to the Wazuh server or manager.
- **Sysmon:** Sysmon plays the role of a monitor because it provides visibility into Windows events, like **process creation**, which **network connection** was made, etc.
- **Invoke Atomic:** Invoke Atomic is an attacking script. It will run them and check if our Wazuh is detecting them or not

Components of ATT&CK Simulation in Wazuh



2- Implementation process

First: check if Sysmon is working on your Wazuh agent or your target machine.

Do: go to Search in Windows machine and type Event Viewer → Application and Services Logs → select Microsoft → Windows → scroll down to Sysmon → operational (this will tell us whether our Sysmon is working or not.)

Secondly: check the Sysmon and PowerShell configuration block in wazuh manager

< agent.conf of Windows_Group

Do: go to Agents management → Group (here we are working with windows group) → open configuration file (agent.conf). So, you can see that we have enabled Sysmon and PowerShell.

If not past this:

```
<!-- Enable Sysmon Logs -->
<localfile>
  <location>Microsoft-Windows-Sysmon/Operational</location>
  <log_format>eventchannel</log_format>
</localfile>
<!-- Enable PowerShell Logs -->
<localfile>
  <location>Microsoft-Windows-PowerShell/Operational</location>
  <log_format>eventchannel</log_format>
</localfile>
```

Save and restart wazuh manager.

Thirdly: install the attacker tool on your Windows agent, which is a powerful framework,

NB: To perform that, we will temporally turn off our antivirus to bypass our execution policy so that we can run the scripts.

Do: run PowerShell as administrator and enter this command: `Set-ExecutionPolicy Bypass CurrentUser` and hit Enter.

We will see this message:

```
Execution Policy Change
The execution policy helps protect you from scripts that you do not trust. Changing the execution policy
might expose you to the security risks described in the execution policy?
[Y] Yes [A] Yes to All [N] No [L] No to All [S] Suspend [?] Help (default is "N"):
Press A. to complete the installation.
```

Fourthly: install Atomic Red Team.

Do: type this command:

```
IEX (IWR 'https://raw.githubusercontent.com/redcanaryco/invoke-atomicredteam/master/install-atomicredteam.ps1' -UseBasicParsing);
```

and hit Enter.

NB: This process will take some time because it will download all the attacking techniques.

After you will see this:

```
NuGet provider is required to continue
PowerShellGet requires NuGet provider version '2.8.5.201' or newer to interact with NuGet-based
repositories. The NuGet provider must be available in 'C:\Program
can also install the NuGet provider by running 'Install-PackageProvider -Name NuGet -MinimumVersion
2.8.5.201 -Force'. Do you want PowerShellGet to install an
[Y] Yes [N] No [S] Suspend [?] Help (default is "Y"):
```

Press Y and hit Enter to complete the installation process.

This:

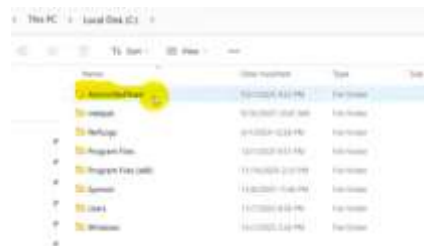
Installation of Invoke-AtomicRedTeam is complete. You can now use the Invoke-AtomicTest function See Wiki at <https://github.com/redcanaryco/invoke-atomicredteam/wiki> for complete details
PS C:\Windows\system32>

Demonstrate that the Atomic Red Team framework has been downloaded.

3- Simulation of attack MITRE T1003 (an OS credential dumping technique)

The attack we are going to simulate here is MITRE T1003 (visit this link: <https://attack.mitre.org/techniques/T1003/> for documentation), which is an OS credential dumping technique. This is hackers' favorite method for stealing passwords.

NB: The path for the Atomic Red Team framework in the C drive.



Copy this.



After that, go to PowerShell, type: cd in it, hit enter, do ls, hit enter to have this:

```
PS C:\Windows\system32> cd C:\AtomicRedTeam
PS C:\AtomicRedTeam> ls

Directory: C:\AtomicRedTeam

Mode                LastWriteTime         Length Name
----                -
d-----          12/1/2025   9:32 PM             atomics
d-----          12/1/2025   9:31 PM      invoke-atomicredteam

PS C:\AtomicRedTeam>
```

Now type: **Invoke-AtomicTest T1003.001 -ShowDetails** and hit enter.

Then, our attack will start.

After that type this: `Invoke-AtomicTest T1003.001` and hit enter

NB: The screen will jump the OS credential because scripts are running.

After you will see this:

```
Exit code: 1
Done executing test: T1003.001-6 Offline Credential Theft With Mimikatz
Executing test: T1003.001-7 LSASS read with pypykatz
The system cannot find the path specified.
Exit code: 1
Done executing test: T1003.001-7 LSASS read with pypykatz
Executing test: T1003.001-8 Dump LSASS.exe Memory using Out-Minidump.ps1
Directory: C:\Users\Usman\AppData\Local\Temp
Mode                LastWriteTime         Length Name
----                -
-a----             12/1/2025   10:20 PM           82977384 lsass_824.dmp
Exit code: 0
```

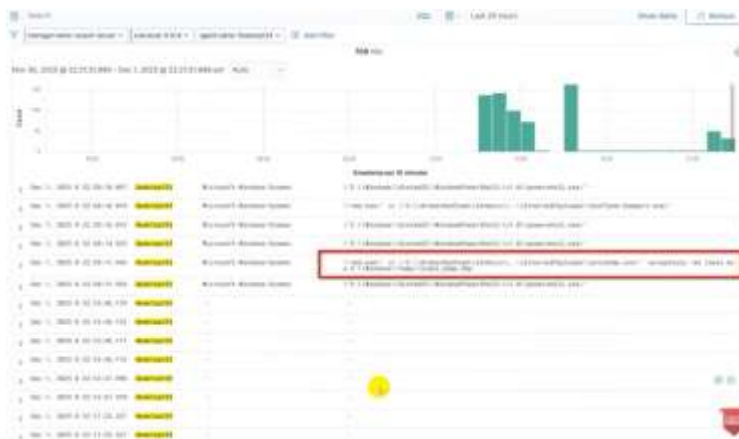
Go to our wazuh SIEM to check logs it generates.

Do: click on severity → to see the logs started coming in.

From there, perform some filter like:

choose the agent (desktop), After that take the decoder name (data.win.system.providerName), And command (data.win.eventdata.parentCommandLine).

Then, we can see: `lsass.exe lsass_dump.dmp` shown to us. The scripts that are currently running are being shown to us here.



We can also check this in another way.

Do: Threat intelligence → MITRE ATT&CK → events to have this:



Alert Name	Agent Name	Rule ID	Rule Meta	Rule Description	Rule Level	Rule ID
Dec 1, 2023 @ 22:05:18:199	Swing123	T1003	Discovery	Powercat executing process discovery	5	010119
Dec 1, 2023 @ 22:05:18:200	Swing123	T1003	Discovery	Powercat Windows cred state execution	5	010120
Dec 1, 2023 @ 22:05:18:201	Swing123	T1003	Discovery	Powercat Windows cred state execution	5	010121
Dec 1, 2023 @ 22:05:18:202	Swing123	T1003	Discovery	Powercat Windows cred state execution	5	010122
Dec 1, 2023 @ 22:05:18:203	Swing123	T1003	Discovery	Powercat Windows cred state execution	5	010123
Dec 1, 2023 @ 22:05:18:204	Swing123	T1003	Discovery	Powercat Windows cred state execution	5	010124
Dec 1, 2023 @ 22:05:18:205	Swing123	T1003	Discovery	Powercat Windows cred state execution	5	010125
Dec 1, 2023 @ 22:05:18:206	Swing123	T1003	Discovery	Powercat Windows cred state execution	5	010126
Dec 1, 2023 @ 22:05:18:207	Swing123	T1003	Discovery	Powercat Windows cred state execution	5	010127
Dec 1, 2023 @ 22:05:18:208	Swing123	T1003	Discovery	Powercat Windows cred state execution	5	010128
Dec 1, 2023 @ 22:05:18:209	Swing123	T1003	Discovery	Powercat Windows cred state execution	5	010129
Dec 1, 2023 @ 22:05:18:210	Swing123	T1003	Discovery	Powercat Windows cred state execution	5	010130
Dec 1, 2023 @ 22:05:18:211	Swing123	T1003	Discovery	Powercat Windows cred state execution	5	010131
Dec 1, 2023 @ 22:05:18:212	Swing123	T1003	Discovery	Powercat Windows cred state execution	5	010132
Dec 1, 2023 @ 22:05:18:213	Swing123	T1003	Discovery	Powercat Windows cred state execution	5	010133
Dec 1, 2023 @ 22:05:18:214	Swing123	T1003	Discovery	Powercat Windows cred state execution	5	010134
Dec 1, 2023 @ 22:05:18:215	Swing123	T1003	Discovery	Powercat Windows cred state execution	5	010135
Dec 1, 2023 @ 22:05:18:216	Swing123	T1003	Discovery	Powercat Windows cred state execution	5	010136
Dec 1, 2023 @ 22:05:18:217	Swing123	T1003	Discovery	Powercat Windows cred state execution	5	010137
Dec 1, 2023 @ 22:05:18:218	Swing123	T1003	Discovery	Powercat Windows cred state execution	5	010138
Dec 1, 2023 @ 22:05:18:219	Swing123	T1003	Discovery	Powercat Windows cred state execution	5	010139
Dec 1, 2023 @ 22:05:18:220	Swing123	T1003	Discovery	Powercat Windows cred state execution	5	010140

The Role of lsass.exe (Local Security Authority Subsystem Service)

- **What it is:** lsass.exe is a critical, core system process built into the Microsoft Windows operating system.
- **Its Main Job:** It is responsible for enforcing security policies, verifying user logins, handling password changes, and managing access tokens.
- **Why it holds sensitive data:** To provide seamless single sign-on (SSO) and manage network authentication, lsass.exe caches active user credentials in memory. This includes NTLM password hashes and Kerberos tickets.
- **Why attackers target it:** Because all these secrets live inside its memory space, lsass.exe is the primary target for **OS Credential Dumping (MITRE ATT&CK T1003)**. If an adversary can read its memory, they can steal credentials and use them to move laterally across a network.

What is lsass_dump.dmp?

- **Definition:** A .dmp (dump) file is a snapshot of a process's memory at a specific point in time.
- **How it's created during the simulation:** When Invoke Atomic Red Team executed the T1003 test script, it forced a utility (or a Windows API call like MiniDumpWriteDump) to target lsass.exe, freeze or read its memory space, and write the contents out to disk as a file—often named something like lsass_dump.dmp.
- **The Security Implication:** Even though this was done safely inside your lab environment, a real-world attacker creates an lsass_dump.dmp file (or streams its contents covertly) so they can take that file offline, run offline password crackers or parsers against it, and extract cleartext or hashed passwords belonging to domain administrators and other users.

Why Wazuh and Sysmon Flagged This

Because lsass.exe is so heavily targeted, security tools monitor it very closely:

1. **Sysmon (Event ID 10):** Detects when an unauthorized or unusual tool attempts to open a "handle" to lsass.exe to read its memory.
2. **Wazuh:** Correlates that Sysmon event, maps it to the **MITRE ATT&CK T1003** framework, and flags the command parameters showing that an attempt was made to extract OS credentials.



WAZUH
Open Source Security
Monitoring Platform

Simulating MITRE ATT&CK with Invoke-Atomic Red Team in Wazuh

Test • Detect • Hunt • Respond

**MITRE
ATT&CK**

Know the Adversary.
Stop the Attack.

1 What is MITRE ATT&CK?



MITRE ATT&CK is a global open knowledge base that describes tactics, techniques and procedures (TTPs) used by attackers in real-world environments.

The 3 key components (TTPs):



Tactics - Why? (attacker's purpose)



Techniques - How? (methods used)



Procedures - Specific details of the attack.

2 The 3 Main Components in This Process



Wazuh Agent
(collects logs)



Sysmon
(monitors events)



Invoke Atomic
(attacking script)

Wazuh Agent

- Collects logs from the endpoint
- Sends to Wazuh manager (Data transmission)

Sysmon

- Monitors Windows events
- Process creation, network connections, etc.

Invoke Atomic

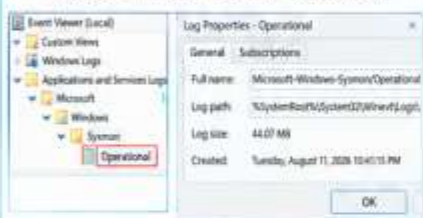
- Runs attack simulations
- Checks if Wazuh detects them

3 Implementation Process

- 1 Check if Sysmon is working (Event Viewer).
- 2 Verify Sysmon & PowerShell config in Wazuh (agent.conf).
- 3 Install Atomic Red Team (turn off antivirus, set execution policy).
- 4 Download and install Invoke-AtomicRedTeam.
- 5 Simulate attack (MITRE T1003).
- 6 Check logs in Wazuh (alerts & filters).

4 Check if Sysmon is Working

- 1 Open Event Viewer → Application and Services Logs → Microsoft → Windows → Sysmon → Operational.



✓ This confirms that Sysmon is collecting logs.

5 Verify Sysmon & PowerShell Config in Wazuh

- 2 Go to Agent Management → Groups → Windows → open agent.conf and check the configuration blocks.

```
<!-- Enable Sysmon Logs -->
<localfile>
  <location>Microsoft-Windows-Sysmon/Operational</location>
  <log_format>eventchannel</log_format>
</localfile>

<!-- Enable PowerShell Logs -->
<localfile>
  <location>Microsoft-Windows-PowerShell/Operational</location>
  <log_format>eventchannel</log_format>
</localfile>
```

✓ Save the configuration and restart the Wazuh manager.

6 Install Invoke-AtomicRedTeam

- 1 Turn off antivirus (temporary).
 - 2 Set execution policy:


```
Set-ExecutionPolicy Bypass -CurrentUser
```
 - 3 Install Atomic Red Team:


```
curl -O "https://raw.githubusercontent.com/redcanaryco/Invoke-AtomicRedTeam/master/install-atomicredteam.ps1" -UseBasicParsing;
```
- Net provider is required to continue...
- [Y] Yes [N] No [S] Suspend [?] Help (default is "Y"): **Y**
- ✓ Installation complete.

7 Simulate Attack (MITRE T1003)

- 2 Run the Atomic Red Team test:

Invoke-AtomicTest T1003.001 -ShowDetails

```
Invoke-AtomicTest T1003.001
Executing test: T1003.001
Collecting LSASS memory
...
PS C:\> Invoke-AtomicTest T1003.001
[+] Starting test ...
[+] Accessing LSASS ...
[+] Dumping credentials ...
[+] Test completed.
```

MITRE ATT&CK T1003 (OS Credential Dumping)

Then run the test:

Invoke-AtomicTest T1003.001

8 Check Logs in Wazuh

- 2 Go to Wazuh dashboard → Severity → filter your agent (Windows desktop) and use these filters:

Severity: Windows desktop

data.win.sysmon.poline

data.win.eventdata.parentCommandline

```
"rule.mitre.id": "T1003",
"rule.mitre.tactic": "Credential-access",
"rule.mitre.technique": "OS Credential Dumping",
"data.win.eventdata.parentCommandline": "C:\Windows\system32\lsass.exe",
"data.win.system.providerName": "Microsoft-Windows-Sysmon"
```

✓ You will see logs like lsass.exe and lsass_dump.dmp.

9 Alternative View in Wazuh

- 2 Go to Threat Intelligence → MITRE ATT&CK → Events. Check the details:



✓ You can also search the technique ID on the MITRE ATT&CK website to get more details.

10 Success! SIEM Telemetry Pipeline is Operational



Windows Endpoint
(Sysmon + PowerShell)



Wazuh Agent
(collects & sends logs)



Wazuh Manager
(analyzes & correlates)



MITRE ATT&CK
(maps tactics & techniques)

Key Benefits

- ✓ Detects fileless malware (PowerShell, memory).
- ✓ Maps to MITRE ATT&CK tactics and techniques.
- ✓ Provides real-time alerts and rich context.
- ✓ Improves threat hunting and incident response.



WAZUH
Open Source Security
Monitoring Platform



Detect



Investigate



Respond



Hunt



Author:
Tims TCHOUAMOU

"Knowledge
Builds a Safer
Tomorrow"